

Digital Nurture SL4J Logging exercises HandsOn

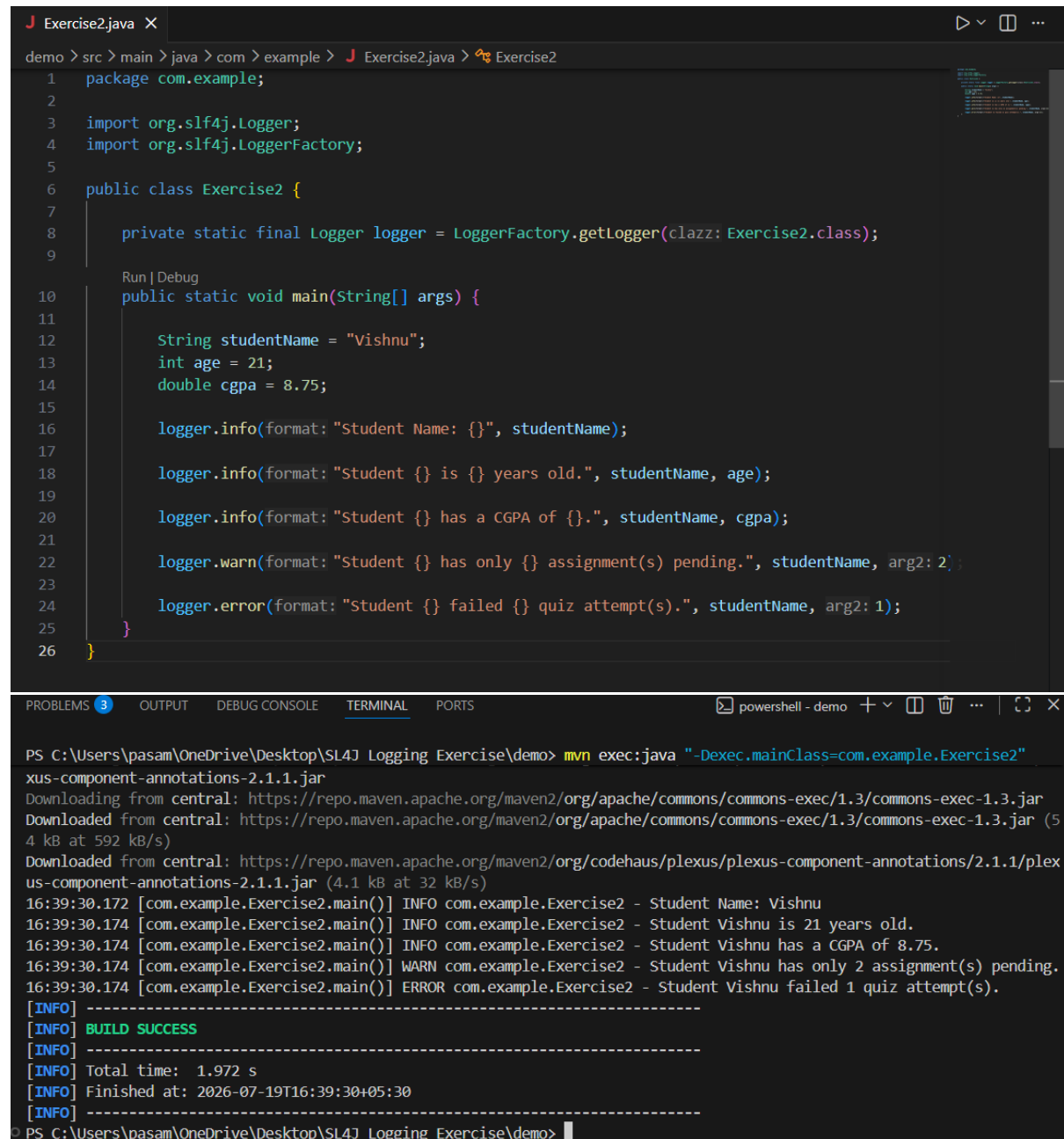
Exercise 1: Logging Error Messages and Warning Levels

```
J App.java X
demo > src > main > java > com > example > J App.java > App
1 package com.example;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5
6 public class App {
7
8     // Create a logger object
9     private static final Logger logger = LoggerFactory.getLogger(App.class);
10
11     Run | Debug
12     public static void main(String[] args) {
13
14         logger.error("This is an error message");
15
16         logger.warn("This is a warning message");
17     }
18 }
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\pasam\OneDrive\Desktop\SL4J Logging Exercise\demo> mvn test
[INFO] Building demo 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- resources:3.4.0:resources (default-resources) @ demo ---
[INFO] skip non existing resourceDirectory C:\Users\pasam\OneDrive\Desktop\SL4J Logging Exercise\demo\src\main\resources
[INFO] --- compiler:3.15.0:compile (default-compile) @ demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- resources:3.4.0:testResources (default-testResources) @ demo ---
[INFO] skip non existing resourceDirectory C:\Users\pasam\OneDrive\Desktop\SL4J Logging Exercise\demo\src\test\resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- surefire:3.5.4:test (default-test) @ demo ---
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.000 s
[INFO] Finished at: 2026-07-19T16:33:26+05:30
```

Exercise 2: Parameterized Logging



The screenshot displays an IDE with a Java file named `Exercise2.java` and its execution output in the terminal.

Exercise2.java Code:

```
1 package com.example;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5
6 public class Exercise2 {
7
8     private static final Logger logger = LoggerFactory.getLogger(clazz: Exercise2.class);
9
10    Run | Debug
11    public static void main(String[] args) {
12
13        String studentName = "Vishnu";
14        int age = 21;
15        double cgpa = 8.75;
16
17        logger.info(format: "Student Name: {}", studentName);
18
19        logger.info(format: "Student {} is {} years old.", studentName, age);
20
21        logger.info(format: "Student {} has a CGPA of {}.", studentName, cgpa);
22
23        logger.warn(format: "Student {} has only {} assignment(s) pending.", studentName, arg2: 2);
24
25        logger.error(format: "Student {} failed {} quiz attempt(s).", studentName, arg2: 1);
26    }
```

Terminal Output:

```
PS C:\Users\pasam\OneDrive\Desktop\SL4J Logging Exercise\demo> mvn exec:java "-Dexec.mainClass=com.example.Exercise2"
xus-component-annotations-2.1.1.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-exec/1.3/commons-exec-1.3.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-exec/1.3/commons-exec-1.3.jar (5
4 kB at 592 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-component-annotations/2.1.1/plex
us-component-annotations-2.1.1.jar (4.1 kB at 32 kB/s)
16:39:30.172 [com.example.Exercise2.main()] INFO com.example.Exercise2 - Student Name: Vishnu
16:39:30.174 [com.example.Exercise2.main()] INFO com.example.Exercise2 - Student Vishnu is 21 years old.
16:39:30.174 [com.example.Exercise2.main()] INFO com.example.Exercise2 - Student Vishnu has a CGPA of 8.75.
16:39:30.174 [com.example.Exercise2.main()] WARN com.example.Exercise2 - Student Vishnu has only 2 assignment(s) pending.
16:39:30.174 [com.example.Exercise2.main()] ERROR com.example.Exercise2 - Student Vishnu failed 1 quiz attempt(s).
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.972 s
[INFO] Finished at: 2026-07-19T16:39:30+05:30
[INFO] -----
PS C:\Users\pasam\OneDrive\Desktop\SL4J Logging Exercise\demo>
```

Exercise 3: Using Different Appenders

```
logback.xml  Exercise3.java X
demo > src > main > java > com > example > Exercise3.java > Exercise3
1  package com.example;
2
3  import org.slf4j.Logger;
4  import org.slf4j.LoggerFactory;
5
6  public class Exercise3 {
7
8      private static final Logger logger = LoggerFactory.getLogger(clazz: Exercise3.class);
9
10     Run | Debug
11     public static void main(String[] args) {
12
13         logger.trace(msg: "This is a TRACE message.");
14
15         logger.debug(msg: "This is a DEBUG message.");
16
17         logger.info(msg: "This is an INFO message.");
18
19         logger.warn(msg: "This is a WARNING message.");
20
21         logger.error(msg: "This is an ERROR message.");
22     }
23 }
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\pasam\OneDrive\Desktop\SL4J Logging Exercise\demo> mvn exec:java "-Dexec.mainClass=com.example.Exercise3"
[INFO] Scanning for projects...
[INFO] -----< com.example:demo >-----
[INFO] Building demo 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- exec:3.1.0:java (default-cli) @ demo ---
16:43:10.764 [com.example.Exercise3.main()] DEBUG com.example.Exercise3 - This is a DEBUG message.
16:43:10.765 [com.example.Exercise3.main()] INFO com.example.Exercise3 - This is an INFO message.
16:43:10.766 [com.example.Exercise3.main()] WARN com.example.Exercise3 - This is a WARNING message.
16:43:10.766 [com.example.Exercise3.main()] ERROR com.example.Exercise3 - This is an ERROR message.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.606 s
[INFO] Finished at: 2026-07-19T16:43:10+05:30
[INFO] -----
PS C:\Users\pasam\OneDrive\Desktop\SL4J Logging Exercise\demo>
```